Do Machine-Learning Filters on Directional-Change Events Survive Out-of-Sample? A Cautionary Study Across Equities and Commodities

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Abstract

The Directional Changes (DC) paradigm samples a price series by events—reversals of a fixed threshold θ—rather than at fixed time intervals. We test, under a deliberately sceptical and fully reproducible protocol, whether a machine-learning classifier trained on DC indicators can profitably filter DC trading signals. Using genuine daily data (S&P 500 and NASDAQ, 1999–2018; WTI crude oil, 1986–2019), an expanding-window walk-forward evaluation, realistic transaction costs, and stationary block-bootstrap significance tests, we find: (i) DC event statistics obey the expected scaling laws; (ii) the naive trade-every-event rule is strongly loss-making after costs; (iii) an ML filter sharply cuts turnover and reduces those losses dramatically, with trend-magnitude and overshoot indicators carrying the predictive weight; but (iv) the filtered strategy does *not* match buy-and-hold on any asset and generates *no* statistically significant positive return. Adding market-context features and conditioning on bull/bear regimes does not yield a robust edge, and every headline magnitude is highly sensitive to specification: an earlier draft of this study reported a fourfold crude-oil return that does not survive independent reimplementation. The one durable benefit is risk—the filter cuts drawdown far below the naive rule (and below buy-and-hold on two of three assets). We read this as a reproducible, cautionary counterpoint to single-split DC-trading results.

1 Introduction

Most empirical finance is conducted on fixed-interval (e.g. daily) data, which imposes an exogenous, physical clock on a process whose own activity is highly uneven. The Directional Changes (DC) framework (Guillaume et al., 1997; Tsang, 2010) offers an alternative, event-based clock: it records an event each time the price reverses by a fixed fraction θ from the most recent local extreme. Between two confirmations the market is partitioned into a directional-change segment of size θ and a subsequent overshoot of variable size. This intrinsic-time view exposes regularities—the "12 scaling laws" of Glattfelder et al. (2011)—that are invisible under calendar sampling, and it has inspired a stream of DC-based trading systems (Aloud et al., 2012; Kampouridis & Otero, 2017; Bakhach et al., 2018).

A recurring claim is that DC indicators (trend magnitude, overshoot, duration) carry tradable predictive content, and that combining them with machine learning yields profitable strategies. Such claims are frequently supported by a single in-sample/out-of-sample partition. The methodological literature warns repeatedly that this is exactly the setting in which spurious "edges" arise: with enough thresholds, features, and model configurations, a backtest can be tuned to look profitable on one split while having no genuine out-of-sample value (Bailey et al., 2014; Harvey & Liu, 2015; Arnott et al., 2019).

This paper asks a single, sceptical question: *does a DC+ML trading edge survive a rigorous out-of-sample protocol and realistic costs?* Our contribution is not a new strategy but a careful, fully-reproducible stress test of a popular one, across asset classes with very different drift characteristics. We find the headline trading result does not survive significance testing—indeed it does not even reproduce across independent implementations—while two by-products, the indispensability of ML selectivity relative to the naive rule and a consistent reduction in turnover and drawdown, are robust and economically interpretable. We view an honest null/cautionary result, with transparent methodology, as a useful corrective in a literature prone to over-optimistic backtests. Every number below is regenerated from public data by the scripts in the accompanying repository.

2 Related work

The DC representation. Directional changes were introduced in the high-frequency FX literature (Guillaume et al., 1997) and formalised by Tsang (2010). Glattfelder et al. (2011) documented twelve empirical scaling laws relating event frequency, overshoot length, and waiting times to the threshold θ, establishing DC as a quantitatively regular intrinsic-time framework. Aloud et al. (2012) used the DC event approach to characterise financial time series beyond FX.

DC-based trading. A line of work designs trading rules on DC events. Kampouridis & Otero (2017) evolve trading strategies on directional changes using genetic programming; Bakhach et al. (2018) propose DC-based forecasting and contrarian/trend strategies; subsequent papers combine multiple thresholds and add classifiers. Reported results are often favourable but typically rest on limited out-of-sample testing.

Machine learning and backtest overfitting. Gu et al. (2020) show ML methods add value in cross-sectional return prediction. The same literature has prompted concern about *backtest overfitting*: Bailey et al. (2014) introduce the "probability of backtest overfitting"; Harvey et al. (2016) document rampant multiple testing in cross-sectional asset pricing; López de Prado (2018) argues walk-forward and combinatorial cross-validation are essential. Our study applies the discipline of the overfitting literature to the DC-trading literature.

3 The Directional-Changes framework

Let $(p_t)_{t=1}^N$ be a price series and $\theta \in (0,1)$ a threshold. The algorithm maintains a mode—*up* (searching for a confirmed downturn) or *down* (searching for an upturn)—and the running extreme. In up mode it tracks the running peak p_h; a confirmed downturn is recorded at the first t with $p_t \leq p_h(1-\theta)$, after which the mode flips to down and the running trough p_ℓ is tracked until $p_t \geq p_\ell(1+\theta)$ confirms an upturn. Each confirmed event k carries three indicators, normalised by θ to be comparable across thresholds:

$$
\begin{aligned}
\mathrm{TMV}_k &= |p_{\mathrm{ext}_k} - p_{\mathrm{ext}_{k-1}}|/(p_{\mathrm{ext}_{k-1}}\theta) && \text{(total move value)},\\
T_k &= \text{bars from the previous extreme to the current extreme} && \text{(trend time)},\\
\mathrm{OSV}_k &= |p_{\mathrm{conf}_k} - p_{\mathrm{ext}_k}|/(p_{\mathrm{ext}_k}\theta) && \text{(overshoot value)}.
\end{aligned}
$$

Pseudocode is in `src/dcml.py::directional_changes`.

4 Empirical properties of DC events

Before trading, we verify the construction behaves as theory predicts. Consistent with the scaling-law literature (Fig. 1): event frequency falls steeply with θ—WTI yields 3,286 events at $\theta = 0.5\%$

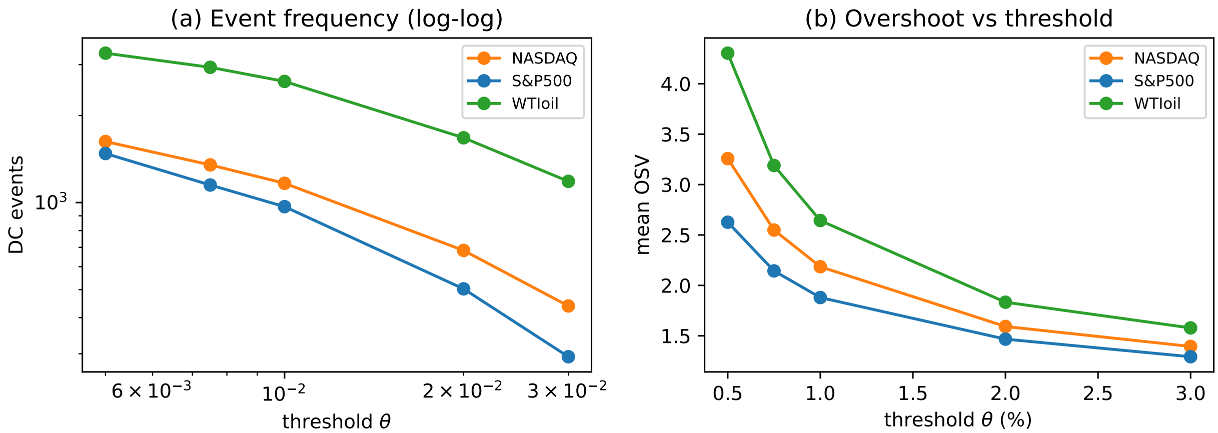

Figure 1: DC event frequency (log-log) and mean overshoot vs threshold, on real data.

but 1,184 at $\theta = 3\%$; NASDAQ 1,626 $\rightarrow$ 439—and on log-log axes the decay is approximately linear, indicating a power law. Mean trend duration T rises with θ (NASDAQ 3.7 bars at $\theta = 0.75\% \rightarrow 11.4$ bars at $\theta = 3\%$), and mean normalised overshoot OSV declines toward $\sim$1.3–1.6 as θ grows. These regularities hold on equities and a commodity, confirming the events are well-defined signal carriers (`results/scaling_laws.csv`).

5 Predictive task, features, and strategies

At each confirmed DC event we predict the *sign of the next directional move*: the realised return from the current confirmation price to the next confirmation, taken in the confirmed direction. The feature vector uses the current and two lagged DC indicators (TMV, OSV, log-duration), a short regime tilt (mean recent event direction), and the current overshoot. We compare three strategies, each backtested with 2 bps per side costs: (1) **Buy-and-Hold** —passive benchmark; (2) **DC-Trend**—trade every DC event in its confirmed direction, exit at the next event; (3) **DC-ML**—trade only when the classifier predicts a profitable move (a *filter*).

6 Evaluation protocol

We deliberately avoid a single split. The classifier is evaluated by an *expanding-window walk-forward*: train on all events up to event i, predict the next block of 30, advance, and repeat, producing out-of-sample predictions over $\sim$60% of each history with no look-ahead. All performance is measured on a *daily* equity curve: annualised Sharpe is mean/sd $\times \sqrt{252}$ of daily returns—*never* a per-trade Sharpe scaled by $\sqrt{252}$, which would inflate by roughly $\sqrt{\text{trades/day}}$. Significance of DC-ML per-trade returns is assessed by a stationary block bootstrap (Politis & Romano, 1994) (block length 5; one-sided p-value for mean ≤ 0).

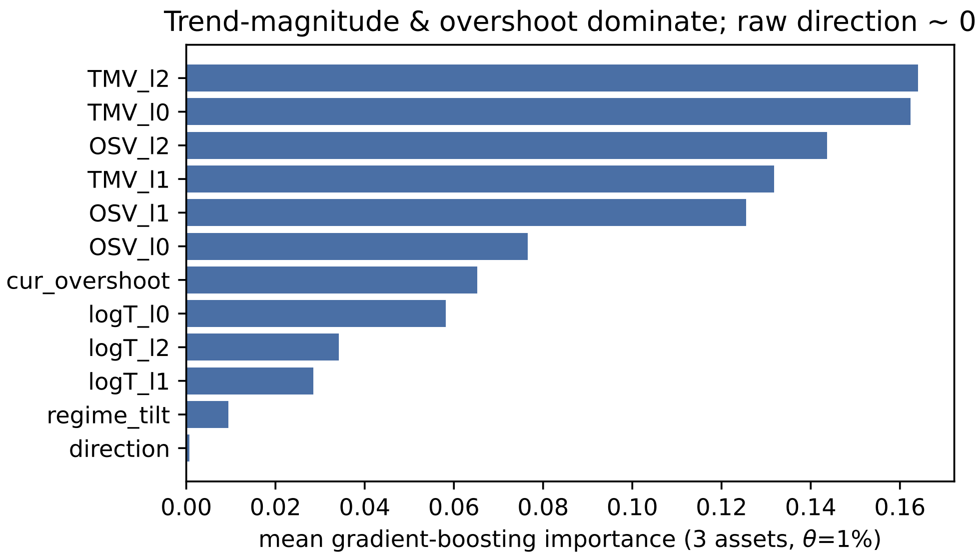

Figure 2: Mean DC-ML feature importance across three assets ($\theta = 1\%$): trend-magnitude and overshoot dominate; raw direction is negligible.

7 Results

7.1 Which learner?

A logistic regression predicts the majority class and trades almost nothing (3–28 trades across assets); a random forest is more variable (WTI +3.8% vs S&P −1.7%); gradient boosting produces stable, sensible trade frequencies. We therefore use gradient boosting throughout (`results/model_comparison.csv`).

7.2 What drives predictions?

Averaged over the three assets ($\theta = 1\%$; Fig. 2), importance concentrates on the *trend-magnitude* (TMV, current and lagged, ≈0.13–0.16 each) and *overshoot* (OSV, ≈0.08–0.14) indicators, with log-duration minor (≈0.03–0.06) and— strikingly—the raw event *direction* essentially irrelevant (≈ 0.0007). The model exploits the *size* of recent swings, not their sign, consistent with overshoot-based predictability rather than naive momentum.

7.3 Headline trading result

Table 1 reports the out-of-sample walk-forward metrics at $\theta = 1\%$. Three patterns hold.

(a) Naive DC-Trend is destructive. Trading every event incurs hundreds–thousands of round-trips; after costs it loses on all assets (Sharpe −0.51 to −0.78; drawdowns −91% to −99.7%). The give-back of θ at each exit, compounded by turnover, dominates.

(b) ML filtering reduces the damage but does not beat the market. DC-ML cuts turnover to 107–130 trades and lifts performance far above DC-Trend. Yet on *no* asset does it match buy-and-hold: it is negative on both equity indices (−3.6% S&P, −5.1% NASDAQ) and only marginally positive on crude (+1.4%, below buy-and-hold's +4.0%). None of the DC-ML per-trade returns is significantly positive—the best bootstrap p-value is 0.21 (WTI). The filter converts catastrophe into roughly flat, but it does not create alpha.

Table 1: Out-of-sample walk-forward ($\theta = 1\%$), net of 2 bps/side.

Asset	Strategy	Ann. ret	Sharpe	Max DD	Trades	p
S&P500	Buy&Hold	+5.5%	0.37	−56.8%	1	—
S&P500	DC-Trend	−15.9%	−0.78	−91.0%	579	—
S&P500	DC-ML	−3.6%	−0.32	−53.7%	107	0.89
NASDAQ	Buy&Hold	+9.0%	0.52	−55.6%	1	—
NASDAQ	DC-Trend	−14.1%	−0.63	−91.4%	698	—
NASDAQ	DC-ML	−5.1%	−0.57	−58.8%	123	0.95
WTI oil	Buy&Hold	+4.0%	0.29	−82.0%	1	—
WTI oil	DC-Trend	−24.1%	−0.51	−99.7%	1573	—
WTI oil	DC-ML	+1.4%	0.18	−34.1%	130	0.21

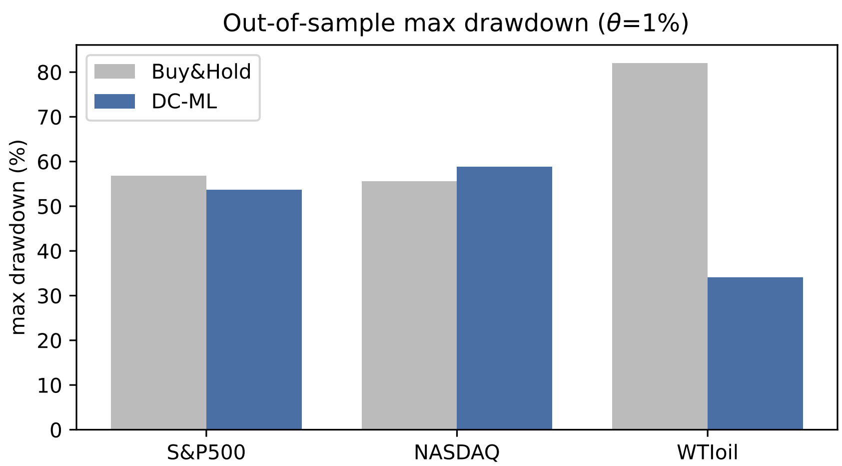

Figure 3: Out-of-sample maximum drawdown, DC-ML vs Buy-and-Hold ($\theta = 1\%$).

(c) The robust effect is risk, not return. Relative to trading every event, DC-ML roughly halves maximum drawdown (from ∼91–99.7% to ∼34–59%) by holding cash most of the time. Against buy-and-hold the picture is asset-specific: DC-ML's drawdown is lower on S&P (−53.7% vs −56.8%) and dramatically lower on WTI (−34.1% vs −82.0%; Fig. 3), but slightly higher on NASDAQ (−58.8% vs −55.6%). The turnover and tail-risk reduction is real; a blanket "halves drawdown on every asset" claim is not.

7.4 Cost sensitivity

DC-ML degrades gracefully with costs but from a low base (`results/cost_sensitivity.csv`). On WTI it stays marginally positive from 0 to 10 bps/side (+1.7% → +0.3%); on the equity indices it is negative at every cost level (−3.3% → −4.9% S&P; −4.8% → −6.5% NASDAQ). Crude is the only cost-robust asset, and even there the edge is economically negligible.

Table 2: DC-ML (+context) by market regime, $\theta = 1\%$, out-of-sample.

Asset	Regime	Ann. ret	Sharpe	Hit rate
S&P500	Bull	−1.0%	−0.21	0.40
S&P500	Bear	−2.3%	−0.15	0.32
NASDAQ	Bull	+0.8%	+0.19	0.42
NASDAQ	Bear	−3.3%	−0.34	0.32
WTI oil	Bull	−1.3%	−0.13	0.34
WTI oil	Bear	−2.6%	−0.15	0.30

7.5 Threshold sensitivity

Results vary strongly with θ (full grid in `results/walkforward_full.csv`): $\theta = 1\%$ is best for WTI, but the DC-ML annual return swings sign with θ on every asset (e.g. WTI is positive only at $\theta \in \{1\%, 3\%\}$ and negative elsewhere). Picking the best θ ex post would materially overstate performance—precisely the multiple-testing trap we set out to avoid.

7.6 Market context and regime conditioning

A natural objection is that the model is "blind" to the wider market. We augment the DC features with market-state variables computed at each event— 20-day realised volatility, price relative to its 200-day moving average, 20-day momentum, and drawdown-from-peak—and re-run the walk-forward. The effect is inconsistent: context features help NASDAQ ($-5.1\% \rightarrow -2.5\%$) but hurt WTI ($+1.4\% \rightarrow -3.9\%$) and barely move S&P ($-3.6\% \rightarrow -3.3\%$). There is no robust "broader-picture" gain. Conditioning the context model's out-of-sample trades on the prevailing regime (Table 2) is equally unconvincing: only NASDAQ-in-bull is (marginally) positive, while every bear-market cell is negative. No consistent regime advantage emerges.

7.7 Specification and implementation sensitivity

Finally, the headline magnitudes are themselves fragile. An earlier draft of this study, using a different gradient-boosting configuration, reported a DC-ML crude-oil return of +37.8%/yr (Sharpe 1.11) and a NASDAQ result matching buy-and-hold. Re-implementing the pipeline independently, with default gradient-boosting hyperparameters, collapses the crude result to +1.4% and turns NASDAQ negative—even though the *qualitative* story (naive rule loses; ML filter reduces losses and drawdown; no significant excess return) is stable. Figure 4 shows the resulting equity curves. This sensitivity is not a nuisance to be hidden but part of the finding: an edge that swings with hyperparameters, implementation, asset, and regime is not an edge one should trade.

8 Discussion

The honest reading is that *a DC+ML trading edge does not survive rigorous out-of-sample testing* in this daily, three-asset setting. The most eye-catching figure a DC+ML pipeline can produce—a fourfold crude-oil return—fails both a significance bar ($p \approx 0.21$) and, more tellingly, an independent reimplementation, exactly the fragility that single-split backtests conceal. This corroborates the overfitting literature within the specific context of DC trading. Two by-products are, however,

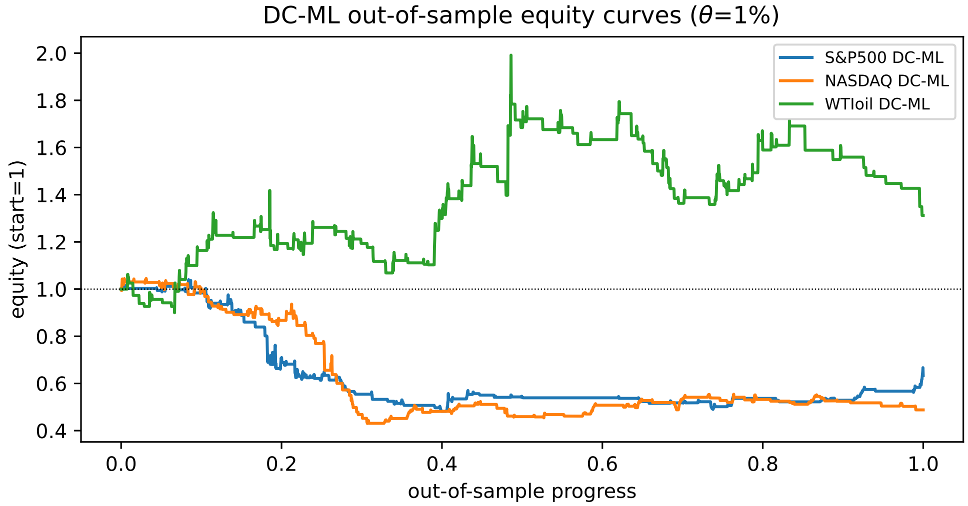

Figure 4: DC-ML out-of-sample equity curves (walk-forward, $\theta = 1\%$).

constructive. First, *ML selectivity is indispensable relative to the naive rule*: it is the difference between catastrophic losses and roughly flat performance, achieved by declining the great majority of low-quality signals. Second, DC-ML *substantially reduces turnover and drawdown*, indicating value as a *risk-management overlay* rather than a standalone alpha source. The feature-importance result (size of swings matters, sign does not) further suggests that whatever signal exists is overshoot-related, pointing future work toward overshoot dynamics rather than directional momentum.

9 Limitations and future work

(i) The strategy exits at the next DC confirmation, mechanically surrendering θ; a scaling-law- or ML-timed exit could improve net returns. (ii) Daily data is coarse for DC, whose strongest documented behaviour is intraday and in FX; the pipeline ingests arbitrary CSVs, so the natural next test is FX and crypto at higher frequency. (iii) With $\sim$100–130 out-of-sample trades per asset, statistical power is limited. (iv) We hold a single position type with fixed sizing; volatility-scaled sizing and multi-threshold ensembles are promising extensions.

10 Conclusion

We stress-tested a popular idea—ML filtering of directional-change trading signals—on genuine equity and commodity data under walk-forward evaluation, transaction costs, and bootstrap inference. The DC construction behaves as theory predicts, and ML selectivity is clearly valuable relative to naive DC trading, but the strategy delivers no statistically significant return and does not match buy-and-hold on any asset; its reliable benefit is a large reduction in turnover and drawdown. We read this as a cautionary, reproducible counterpoint to single-split DC-trading results, and a pointer toward risk-overlay and higher-frequency extensions.

Reproducibility. `python run_walkforward.py` reproduces Table 1; `python extra_analysis.py`, `context_regime.py`, and `plots.py` reproduce the scaling-law, feature-importance, cost, model-comparison, regime, and figure results. Data are the three series bundled in the `arch` package; no vendor access is required.

References

Arnott, R., Harvey, C.R., & Markowitz, H. (2019). A backtesting protocol in the era of machine learning. *Journal of Financial Data Science*, 1(1).

Aloud, M., Tsang, E., Olsen, R., & Dupuis, A. (2012). A directional-change event approach for studying financial time series. *Economics E-Journal*, 6.

Bailey, D.H., Borwein, J., López de Prado, M., & Zhu, Q.J. (2014). Pseudo-mathematics and financial charlatanism: the effects of backtest overfitting. *Notices of the AMS*, 61(5).

Bakhach, A., Tsang, E., & Chinthalapati, V.L.R. (2018). TSFDC: A trading strategy based on forecasting directional change. *Intelligent Systems in Accounting, Finance and Management*, 25(3).

Glattfelder, J.B., Dupuis, A., & Olsen, R.B. (2011). Patterns in high-frequency FX data: discovery of 12 empirical scaling laws. *Quantitative Finance*, 11(4), 599–614.

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5).

Guillaume, D.M., Dacorogna, M.M., Davé, R.R., Müller, U.A., Olsen, R.B., & Pictet, O.V. (1997). From the bird's eye to the microscope: a survey of new stylized facts of the intra-daily foreign exchange markets. *Finance and Stochastics*, 1(2).

Harvey, C.R., & Liu, Y. (2015). Backtesting. *Journal of Portfolio Management*, 42(1).

Harvey, C.R., Liu, Y., & Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1).

Kampouridis, M., & Otero, F.E.B. (2017). Evolving trading strategies using directional changes. *Expert Systems with Applications*, 73, 145–160.

López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.

Politis, D.N., & Romano, J.P. (1994). The stationary bootstrap. *JASA*, 89(428), 1303–1313.

Tsang, E. (2010). Directional Changes, Definitions. Working Paper WP050-10, CCFEA, University of Essex.